

Diet supplementation with donkey milk upregulates liver mitochondrial uncoupling, reduces energy efficiency and improves antioxidant and antiinflam...

Dietary PUFA, mainly those of the n-3 family, are known to play essential roles in the maintenance of energy balance and in the reduction of body fat deposition through the upregulation of mitochondrial uncoupling that is the main source of reactive oxygen species. We hypothesized that rat supplementation with raw donkey's milk (DM), characterized by low-fat content and higher n3:n6 ratio, may affect energy balance, lipid metabolism, and prooxidant status as compared to animals treated with cow's milk. In the present study, the effects of drinking raw DM (for 4 weeks) on energy balance, lipid metabolism, antiinflammatory, and antioxidant/detoxifying defences was compared to that produced by rat intake of an iso-energetic amount of raw cow's milk. The hypolipidemic effect produced by DM paralleled with the enhanced mitochondrial activity/proton leakage and with the increased activity or expression of mitochondrial markers namely, carnitine palmitoyl transferase and uncoupling protein 2. The association of decreased energy efficiency with reduced proinflammatory signs (TNF- α and LPS levels) with the significant increase antioxidant (total thiols) and detoxifying enzyme activities (glutathione-S-transferase NADH quinone oxidoreductase) in DM-treated animals, indicated that beneficial effects were attributable, at least in part, to the activation of nuclear factor 2 erythroid-related factor 2 pathway.

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